ESTEBAN E. MORALES ROJAS

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EDUCATION

Master of Materials Science and Engineering Expected January 2027

University of Massachusetts Amherst

Coursework and lab work in organic electronics, thin-film processing, flexible energy devices, and encapsulation of electronic devices.

Bachelor of Science in Electrical electronics engineering · GPA 9.04/10 (≈ 3.9/4.0)

Universidad Nacional Autónoma de México (UNAM), Mexico City. August 2024

Thesis: open-space Michelson interferometer for defect detection on flat reflective surfaces.

TECHNICAL SKILLS

Processes Magnetron sputtering, thermal evaporation / PVD, spin coating, PiCVD barrier layers, plasma and wetting-layer pretreatment, polymer-substrate transfer, planarization layers, 3D-printed process fixtures.

Devices OSC stack design; transparent dielectric–metal–dielectric electrodes (AZO/Ag/AZO, ITO/Ag/ITO); doped ETLs; flexible / encapsulant integration, deposition ratio.

Characterization Solar-simulator J–V, sEQE, PAIOS, UV–Vis, AFM, SEM, profilometry, four-point probe, FTIR, interferometric metrology.

Tools & software Angstrom three-glovebox PVD/sputter platform; Python, MATLAB, LabVIEW, Origin; Fusion 360, FreeCAD, Cura; KiCad, Multisim, Aeres (Angstrom) sputtering and PVD.

EXPERIENCE

Wearable Electronics Lab (WELab), University of Massachusetts Amherst *Amherst, Massachusetts*

Graduate Research Assistant August 2025 – January 2027

- Operated a connected three-glovebox Angstrom PVD/sputtering platform to build photovoltaic stacks; designed 3D-printed process fixtures and characterized devices with solar-simulator J–V, AFM, profilometry, UV–Vis, IR, and four-point probe to link morphology to efficiency.
- Fabricated fullerene-free organic solar cells (NF3000N:NF3000P) with a doped n-type ETL (PTCDA:Mg); tuned ETL thickness, active-layer morphology, and process windows against reference stacks to produce efficient, stable devices on glass.
- Sputtered dielectric–metal–dielectric (DMD) transparent electrodes (AZO/Ag/AZO, ITO/Ag/ITO) to meet transmittance and sheet-conductivity targets for transparent flexible devices; transferred the bottom electrode onto Bomar BR-640D and compared film growth and device performance on glass versus polymer.
- Deposited PiCVD multilayer barriers on calcium/glass and complete photovoltaic devices to lower WVTR and improve stability under water and oxygen exposure.

KAUST Solar Center — Ultrafast Dynamics Group *Thuwal, Saudi Arabia*

Visiting Student Researcher September 2023 – February 2024

- Designed and automated a sensitive EQE (sEQE) measurement bench in LabVIEW, including remote operation.
- Fabricated P3HT:PCBM organic solar cells and characterized devices on a solar simulator and with PAIOS.
- Built a six-transistor GaN-HEMT ultrafast amplifier (~60 ps response, DC–30 GHz) for high-speed optical detection.

ICAT–UNAM, Fiber Optic Sensors Laboratory *Mexico City, Mexico*

Undergraduate Researcher July 2022 – May 2023

- Built an open-space Michelson interferometer to detect defects on flat reflective surfaces and extract refractive indices of transparent solids (B.E.E. thesis; SOMI XXXVII).
- Adapted the same interferometer for small-displacement measurements under mechanical vibration.
- Designed a laser-diode driver with supply regulation and high-voltage protection.

TRAINING & CERTIFICATIONS

- MIT.nano / Fab.nano cleanroom orientation and protocol training.
- UMass Amherst EHS: laboratory safety, fire safety, hazardous waste management; compressed-gas and LN_2 cylinder handling; glovebox inert-atmosphere practice.
- KAUST Solar Center: Photovoltaic Design; workshops in scientific visualization (VisIt), machine learning, deep learning, data visualization, and basic PCB prototyping.